

treatment plan in Windhoek is a direct potable reuse plant that has been operating since 1968 and does not produce a brine due to the choice of treatment options. However, in instances where brine disposal is required at inland sites, this will be incorporated in the design during the feasibility assessment. The costs of brine handling can be significant and have the potential to compromise the project viability. Hence, it is paramount that this is incorporated in the design from the beginning.

Example Project Summary for an Indicative Project

A municipality undertook a feasibility study looking at a potential wastewater industrial reuse project to be implemented as a Public Private Partnership (PPP). For the Municipality, site-specific future climate projections indicate the following for the 2011-2040 period under both a moderate climate change future (RCP 4.5 scenario) and an extreme climate change future (RCP 8.5 scenario) (Table 7).

Table 7: Site-specific future climate projections for the Municipality 2011-2040 under two scenarios

Climatic Variable (Indicator)	RCP 4.5	RCP 8.5
Temperature (annual mean)	+1°C	+1°C
Rainfall (annual mean)	-3%	-5%
Aridity (annual mean)	+2%	+1%
Soil moisture (annual mean)	-5%	-6%
Water discharge (annual mean)	-3%	-4%
Water runoff (annual mean)	-3%	-4%
Length of longest dry spell	A small increase (<5 days a year)	A small increase (<5 days a year)
Number of dry spells a year	A small decrease (<25% change)	A small decrease (<25% change)

Thus, the Municipality is projected to face rising temperatures from climate change. In addition to decreases in rainfall, runoff, and discharge, the rising heat is likely to drive an increase in aridity, leading to somewhat drier conditions (including lower soil moisture).

As a result, water deficits are projected as being:

- 20 -40% deficit under medium population growth scenarios by 2050, and
- >80% high water stress by 2040 as predicted by the WRI Aquaduct Risk Tool.

The feasibility study indicates that there is, on average, at least 82 Ml/d available for tertiary advanced treated at a proposed regional site. The transaction advisor engaged with potential used for the product water and identified a total of five potential end users who could potentially purchase up to 71.7 Ml/d on average of the product water.

A total of 7 potential technical options and configurations were identified and analysed through a comprehensive weighting and assessment system. The option of a new regional treatment (capacity 75 Ml/day) at a new regional site with a pump collection system was selected as the preferred option.

A preferred site for the treatment plant has been selected noting the proximity to off takers as well as a range of environmental considerations. Noting the brine to be produced by the plant the proximity to the ocean for discharge is important. The requirements for this discharge in terms of the National Environmental Management Act (Act 107 of 1998) and the Integrated Coastal Management Act (Act 24 of 2008) have been included in the feasibility study and are central to the Environmental and Social Impact Assessment and associated management plan. A total of 25 ha will be required for the regional treatment facility.

Pilot plant testing has been undertaken and indicated that the effluent can be treated to an acceptable standard by utilising filtration (presumably ultra-filtration) and reverse osmosis processes. Some additional pre-treatment steps will be necessary.

The total capital cost estimate in 2017 (including land costs) amounted to R1 736 209 178. The Unitary Reference Value was estimated at R10.82 / m³.

The potential for biogas generation was also include in the study and it was recommended that thermophilic digestion for CHP engines be implemented as part of the project. It was however stated that the viability of the project was not dependant on the biogas electricity generation. Furthermore, composting of sludge was also included as part of the project to ensure waste beneficiation.

A collection system consisting of five pumpstation and rising mains will be required to collect and transfer the effluent to the regional treatment facility. Preliminary designs of these collection pipelines indicated a total of 31.6 km of pipelines varying from 500mm to 864mm pipelines will be required. A total installed power of 585 kW will be required for pumping.

The regional facility will produce two quality streams. One of the larger users will receive lower quality re-use water directly from the WWTW (30 MI/d). The other users will receive a higher quality water from the tertiary treatment plant (45MI/d). Treatment at the regional facility will take place in three treatment modules as follows:

- 50 MI/d domestic WWTW
- 35 MI/d Industrial treatment work
- 10 MI/d water treatment works for higher quality water

A 50 MI/d Biological Nutrient removal activated sludge plant will treat the raw sewage diverted from the local WWTW. The 35 MI/d industrial treatment plant will treat industrial effluent from the pulp and paper mills using conventional water treatment, plus advanced oxidation and reverse osmosis membrane treatment. The 10 MI/d water treatment plant will treat a portion of the treated effluent from the WWTW to an industrial standard using conventional treatment processes (Flocculation – sedimentation -filtration) plus advanced oxidation (Ozonation) and Granular Activated Carbon.

The distribution network to off-take users will be via a gravity driven network. This is made possible by the location of the regional facility. An estimated 27 km of pipelines ranging from 225mm diameter to 762mm diameter pipes will make up the network.

As part of the feasibility study, the feasibility of implementing value adding technology in the form of the potential to convert biogas to energy and the potential to convert sludge to a useable end product were investigated and found to be feasible producing sufficient power to significantly reduce the load drawn from the national grid.

Benefits to Ecosystem Services

Currently, treated effluent from the macerators and WWTWs at the municipality is discharged into nearby streams, natural water courses or directly into the ocean through deep sea outfalls. High volumes of discharge that can create significant pollutant loading into these sources can have an effect on these aquatic water bodies, on coastal systems, and on the composition of the water in these sources. A decrease in the amount of effluent released into surface water sources can go a long way to reducing adverse environmental impact. Additionally, substances and microorganisms that are present in treated effluent can have an adverse impact on stream habitats and wetlands, and on the wildlife that is dependent upon sustained water quality attributes of these environments. The diversion of water from surface water bodies can also have a harmful impact on the flora and fauna of ecosystems that exist within these water bodies. By reducing the diversion of water from surface water bodies as well as reducing the amount of effluent that is emitted into these environments through increased wastewater recycling, this can significantly lessen the deterioration of ecosystem health and hence improved wildlife subsistence in these areas.

For the purposes of the economic appraisal, it is assumed that the implementation of the project shall divert the use and demand for potable water and therefore present an economic cost avoided to the value of R739 million at an 8% social discount rate.

Project Selection Criteria